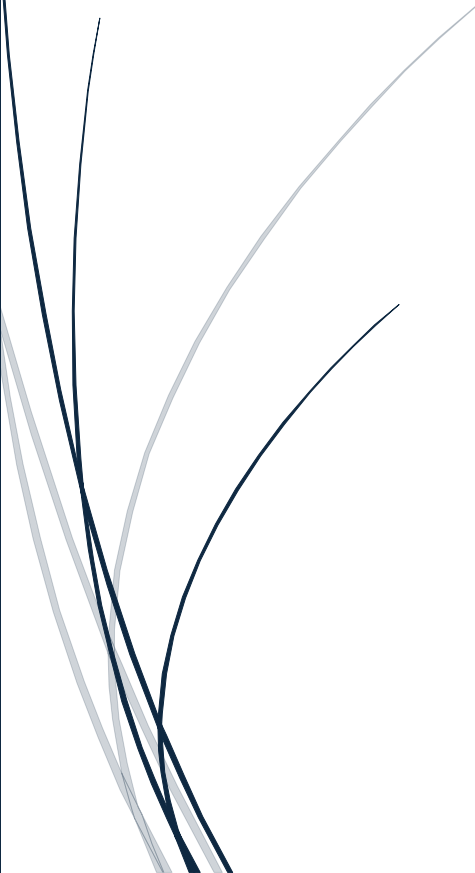




10/15/2025

Week-6 Lab:

Building a Secure VPC Environment
on AWS



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MLEVIN56 ENPM665 0101

Step 1: Create a Custom VPC

The screenshot shows the AWS Management Console for the 'us-east-1' region. The left sidebar contains the 'VPC dashboard' and a list of VPC resources. The main content area displays 'Your VPCs (2)' with a table listing two VPCs:

Name	VPC ID	State	Block Public...	IPv4 CIDR	IPv6 CIDR
-	vpc-02473e2cee2575d1d	Available	Off	172.31.0.0/16	-
MyLabVPC	vpc-08656f8506f5bf1c8	Available	Off	10.0.0.0/16	-

Below the table, there is a section titled 'Select a VPC above'.

Step 2: Create a Public Subnet

The screenshot shows the AWS Management Console for the 'us-east-1' region, specifically the 'Subnets' page. A green notification banner at the top states: 'You have successfully changed subnet settings: Enable auto-assign public IPv4 address'. The main content area displays 'Subnets (1/7)' with a table listing several subnets:

Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR
-	subnet-0e6ec6b894aa3eb40	Available	vpc-02473e2cee2575d1d	Off	172.31.0.0/20
MyLabPublicSubnet	subnet-04d67a545429418db	Available	vpc-08656f8506f5bf1c8 MyLa...	Off	10.0.1.0/24
-	subnet-070503580d6286bba	Available	vpc-02473e2cee2575d1d	Off	172.31.32.0/2
-	subnet-0e391078df5e02c4e	Available	vpc-02473e2cee2575d1d	Off	172.31.48.0/2

Below the table, the details for 'subnet-04d67a545429418db / MyLabPublicSubnet' are shown. The details include:

- Subnet ID:** subnet-04d67a545429418db
- Subnet ARN:** arn:aws:ec2:us-east-1:904727092191:subnet/subnet-04d67a545429418db
- State:** Available
- IPv4 CIDR:** 10.0.1.0/24
- Availability Zone:** us-east-1a
- Available IPv4 addresses:** 251
- Network border group:**
- Block Public Access:** Off
- IPv6 CIDR association ID:**
- Route table:** rtb-0a1e588b92ea1aed5

Step 3: Attach an Internet Gateway & Update Routes

The screenshot displays the AWS Management Console interface for the 'Internet gateways' section. The left-hand navigation pane shows the 'VPC dashboard' with various options like 'Your VPCs', 'Subnets', 'Route tables', and 'Internet gateways' (which is currently selected). The main content area shows a list of internet gateways. One gateway, 'MyLabIGW', is selected, and its details are displayed below the list. The details include the Internet gateway ID, State (Attached), VPC ID, and Owner.

Name	Internet gateway ID	State	VPC ID	Owner
-	igw-075579a6430aad638	Attached	vpc-02473e2cee2575d1d	904727092191
MyLabIGW	igw-06cfa070eb2415069	Attached	vpc-08656f8506f5bf1c8 MyLabVPC	904727092191

igw-06cfa070eb2415069 / MyLabIGW

Details

Internet gateway ID	State	VPC ID	Owner
igw-06cfa070eb2415069	Attached	vpc-08656f8506f5bf1c8 MyLabVPC	904727092191

Root Table:

aws | United States (N. Vi) | Account ID: 9047-2709-2191 | mlevin56-ENPM665-0101-2025

VPC > Route tables

VPC dashboard

AWS Global View

Filter by VPC

▼ **Virtual private cloud**

- Your VPCs
- Subnets
- Route tables**
- Internet gateways
- Egress-only internet gateways
- Carrier gateways
- DHCP option sets
- Elastic IPs
- Managed prefix lists
- NAT gateways
- Peering connections
- Route servers [New](#)

▼ **Security**

- Network ACLs
- Security groups

Route tables (1/2) Info

Last updated less than a minute ago

Actions Create route table

Find route tables by attribute or tag

Name	Route table ID
-	rtb-0a1e588b92ea1ae45

rtb-0a1e588b92ea1ae45

Details Routes Subnet associations

Details

Route table ID
[rtb-0a1e588b92ea1ae45](#)

VPC
[vpc-08656f8506f5bf1c8](#) | MyLabVPC

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7:24 PM 10/25/2025

aws

United States (N. Vi

Account ID: 9047-2709-2191
mlevin56-ENPM665-0101-2025

☰

VPC > Route tables > rtb-0a1e588b92ea1ae45

Info

VPC dashboard

AWS Global View

Filter by VPC

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

Carrier gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Security

Network ACLs

Security groups

Details

Route table ID

rtb-0a1e588b92ea1ae45

VPC

vpc-08656f8506f5bf1c8 | MyLabVPC

Explicit subnet associations

subnet-04d67a545429418db / MyLabPublicSubnet

Main

Yes

Owner ID

904727092191

Edge associations

-

Routes

Subnet associations

Edge associati

Routes (2)

Both

Edit routes

Filter routes

< 1 >

Destination	Target	Status	Propag...
0.0.0.0/0	igw-06cfa0...	Active	No
10.0.0.0/16	local	Active	No

CloudShell

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Step 4: Launch an EC2 Instance

Key pairs | EC2 | us-east-1

us-east-1.console.aws.amazon.com/ec2...

aws

EC2 > Key pairs

Capacity Reservations

Capacity Manager [New](#)

▼ Images

AMIs

AMI Catalog

▼ Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

▼ Network & Security

Security Groups

Elastic IPs

Placement Groups

[Key Pairs](#)

Network Interfaces

▼ Load Balancing

Load Balancers

Target Groups

Trust Stores

Successfully created key pair

Key pairs (1) [Info](#) [Actions](#) [Create key pair](#)

Find Key Pair by attribute or tag

☐	Name	Type	Created	Finger
☐	MyLabKeyPair	rsa	2025/10/25 1...	22:af

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Launch the EC2 Instance

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#InstanceDetails:instanceId=i-0ca6aeaf5e1719f2e

aws

Search

[Alt+S]

United States (N. Virginia)

Account ID: 9047-2709-2191

mleivn56-ENPM665-0101-2025

EC2

Instances

i-0ca6aeaf5e1719f2e

EC2

Dashboard

EC2 Global View

Events

Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Capacity Manager

Images

AMIs

AMI Catalog

Elastic Block Store

Volumes

CloudShell

Feedback

Instance summary for i-0ca6aeaf5e1719f2e (MyTestInstance)

Updated less than a minute ago

Instance ID

i-0ca6aeaf5e1719f2e

Public IPv4 address

98.92.202.81 | open address

Private IPv4 addresses

10.0.1.92

IPv6 address

-

Instance state

Running

Public DNS

-

Hostname type

IP name: ip-10-0-1-92.ec2.internal

Private IP DNS name (IPv4 only)

ip-10-0-1-92.ec2.internal

Elastic IP addresses

-

Answer private resource DNS name

-

Instance type

t3.micro

AWS Compute Optimizer finding

Opt-in to AWS Compute Optimizer for recommendation

Auto-assigned IP address

98.92.202.81 [Public IP]

VPC ID

vpc-08656f8506f5bf1c8 (MyLabVPC)

Learn more

IAM Role

-

Subnet ID

subnet-04d67a545429418db (MyLabPublicSubnet)

Auto Scaling Group name

-

IMDSv2

Required

Instance ARN

arn:aws:ec2:us-east-1:904727092191:instance/i-0ca6aeaf5e1719f2e

Managed

false

CloudShell

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53°F

Mostly cloudy

Search

7:39 PM

10/25/2025

aws

Search

[Alt+S]

United States (N. Virginia)

Account ID: 9047-2709-2191

mleivn56-ENPM665-0101-2025

EC2

Instances

Instances (1/1)

Connect

Instance state

Actions

Launch instances

Find Instance by attribute or tag (case-sensitive)

All states

Name

Instance ID

Instance state

Instance type

Status check

Alarm status

Availability Zone

Public IPv4

MyTestInstance

i-0ca6aeaf5e1719f2e

Running

t3.micro

3/3 checks passed

View alarms

us-east-1a

-

i-0ca6aeaf5e1719f2e (MyTestInstance)

Inbound rules

Filter rules

Name

Security group rule ID

Port range

Protocol

Source

Security groups

-

sgr-03732657860e094bd

22

TCP

216.15.0.122/32

MyLabSG

Outbound rules

CloudShell

Feedback

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Search

8:00 PM

10/25/2025

Step 5. Connect to Your EC2 Instance and Test Connectivity

The screenshot displays the AWS Management Console interface for the EC2 Instances page. The top navigation bar shows the AWS logo, search, and account information (Account ID: 9047-2709-2191, mlevin56-ENPM665-0101-2025). The left sidebar lists navigation options under EC2, including Dashboard, EC2 Global View, Events, Instances, Images, and Elastic Block Store. The main content area shows the 'Instances (1/1)' page with a table of instances. The instance 'i-0ca6aeaf5e1719f2e (MyTestInstance)' is selected, and its details are shown in the 'Details' tab. The instance summary includes the Instance ID, Public IPv4 address (98.92.202.81), Private IPv4 addresses (10.0.1.92), and IPv6 address.

Instances (1/1) Info

Connect Instance state Actions

Launch instances

Find Instance by attribute or tag (case-se.) All states

1

v4 DNS	Public IPv4 ...	Elastic IP	IPv6
	98.92.202.81	-	-

i-0ca6aeaf5e1719f2e (MyTestInstance)

Details Status and alarms Monitoring

Instance summary Info

Instance ID i-0ca6aeaf5e1719f2e	Public IPv4 address 98.92.202.81 open address
Private IPv4 addresses 10.0.1.92	IPv6 address -

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7:40 PM 10/25/2025

Part 2: VPC Flow Logs Configuration

The screenshot displays the AWS CloudWatch console interface. At the top, the navigation bar shows the AWS logo, account ID (9047-2709-2191), and region (United States (N. VI)). The main header indicates the current page is 'Log groups' under the 'CloudWatch' service.

A green notification banner at the top right states: "Log group 'my-flow-logs' has been created." Below this, the 'Log groups (1)' section is visible, featuring a 'Create log group' button and a 'Start tailing' button. A note mentions that by default, only up to 10,000 log groups are loaded.

The left sidebar contains a navigation menu with categories like 'Dashboards', 'AI Operations', 'Alarms', 'Logs', 'Metrics', 'Application Signals (APM)', 'GenAI Observability', and 'Network Monitoring'. The 'Logs' category is expanded, showing 'Log groups' as the selected option.

The main content area shows a search bar for log groups, a filter for 'Exact match', and a table with columns for 'D...', 'S...', 'Retenti...', and 'Metric ...'. The table currently displays one entry with a retention period of '1 month'.

The bottom of the screen shows the Windows taskbar with various application icons and the system clock indicating 8:06 PM on 10/25/2025.

Step 2: Step Create VPC Flow Logs for your VPC or Subnet

The screenshot displays the AWS Management Console interface for a VPC Flow Log. The browser address bar shows the URL: `us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#FlowLogDetails:flowlogId=fl-079142b1b82edf418`. The console header includes the AWS logo, a search bar, and account information for 'United States (N. Virginia)' with Account ID: 9047-2709-2191.

The left-hand navigation pane shows the 'VPC dashboard' with a 'Filter by VPC' dropdown. Under 'Virtual private cloud', 'Your VPCs' is selected. Under 'Security', 'Network ACLs' is visible.

A green success banner at the top states: 'Successfully created flow log for the following resource: vpc-08656f8506f5bf1c8'. Below this, the flow log ID 'fl-079142b1b82edf418' is displayed with an 'Actions' button.

The 'Details' section provides the following information:

Details	
Flow Log ID fl-079142b1b82edf418	Destination Type cloud-watch-logs
Name -	Destination Name my-flow-logs
State Active	IAM Role arn:aws:iam::904727092191:role/service-role/MyVPCLogs
Creation Time Saturday, October 25, 2025 at 20:10:28 EDT	Cross Account IAM Role -
Traffic Type ALL	Max Aggregation Interval 10 minutes
File Format -	Log Format Default
	Hive Compatible Partitions -
	Partition Logs -

Below the details, there are tabs for 'Tags' and 'Integrations'. The 'Tags' tab is active, showing a search bar and a table with columns 'Key' and 'Value'. A 'Manage tags' button is located to the right.

The footer of the console shows the date '© 2025, Amazon Web Services, Inc. or its affiliates.' and links for 'Privacy', 'Terms', and 'Cookie preferences'. The system tray at the bottom indicates a temperature of 53°F and 'Mostly cloudy' weather.

Step 3: Verify the Flow Logs

The screenshot displays the AWS CloudWatch console interface. At the top, the navigation bar shows the AWS logo, account ID (9047-2709-2191), and region (United States (N. Vi)). The breadcrumb trail indicates the path: CloudWatch > Log groups > my-flow-logs. The left-hand navigation pane lists various CloudWatch features, with 'Logs' expanded to show 'Log groups' as the selected option. The main content area is titled 'Log streams (1/1)' and includes buttons for 'Delete' and 'Create log stream', along with a 'Search all log streams' button. A search filter is present with the text 'Filter log streams or try prefix search'. Below the search bar, there are checkboxes for 'Exact match' and 'Show expired', and a pagination control showing '1' of 1 items. A table lists the log streams, with one entry selected: 'eni-09ee9f1b54d81c064-' with a last event time of '2025-10-26 00:11:1...'. The bottom of the screen shows the Windows taskbar with various application icons and the system clock indicating 8:12 PM on 10/25/2025.

Account ID: 9047-2709-2191
mlevin56-ENPM665-0101-2025

CloudWatch > Log groups > my-flow-logs

CloudWatch

Favorites and recents

Dashboards

► AI Operations [New](#)

► Alarms [0](#) [0](#) [0](#)

▼ Logs

Log groups

Log Anomalies

Live Tail

Logs Insights

Contributor Insights

► Metrics [New](#)

► Application Signals (APM) [New](#)

► GenAI Observability [New](#)

► Network Monitoring

Custom field indexes
[Configure](#)

Transformer
[Configure](#)

< Log streams Tags Anomaly detection >

Log streams (1/1)

[Refresh](#) [Delete](#) [Create log stream](#)

[Search all log streams](#)

By default, we only load the most recent log streams.

☐ Exact match ☐ Show expired [Info](#) < 1 > [Settings](#)

☒ Log stream Last event time

☒ eni-09ee9f1b54d81c064- 2025-10-26 00:11:1...

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8:12 PM
10/25/2025

```
[ec2-user@ip-10-0-192 ~]$ ssh-keygen -t rsa -C "root@8.8.8.8"
Generating public/private rsa key pair.
Enter passphrase (empty for no passphrase): 
Enter same passphrase again: 
Your identification has been saved in /home/ec2-user/.ssh/id_rsa.
The authenticity of host '8.8.8.8 (8.8.8.8)' can't be established.
ED25519 key fingerprint is SHA256:mGd3IoicHElJP3uuIcNWkAdnpEbYvR0AnOA9hxxPvr4.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '8.8.8.8' (ED25519) to the list of known hosts.
```

```
#_#
| \ ##### Amazon Linux 2023
~ _#####\
...    |###|\
      \|##|V--> https://aws.amazon.com/linux/amazon-linux-2023
       V---^
     ..../
   ...//
 m/_/
```

```
[ec2-user@ip-10-0-192 ~]$ ping -c 4 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=113 time=1.39 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=113 time=1.35 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=113 time=1.43 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=113 time=1.68 ms

--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 1.353/1.462/1.676/0.126 ms
[ec2-user@ip-10-0-192 ~]$ ping google.com
PING google.com (142.251.167.101) 56(84) bytes of data.
64 bytes from ww-in-f101.1e100.net (142.251.167.101): icmp_seq=1 ttl=102 time=1.86 ms
64 bytes from ww-in-f101.1e100.net (142.251.167.101): icmp_seq=2 ttl=102 time=1.86 ms
64 bytes from ww-in-f101.1e100.net (142.251.167.101): icmp_seq=3 ttl=102 time=1.94 ms
64 bytes from ww-in-f101.1e100.net (142.251.167.101): icmp_seq=4 ttl=102 time=1.86 ms
64 bytes from ww-in-f101.1e100.net (142.251.167.101): icmp_seq=5 ttl=102 time=1.93 ms
64 bytes from ww-in-f101.1e100.net (142.251.167.101): icmp_seq=6 ttl=102 time=1.85 ms
64 bytes from ww-in-f101.1e100.net (142.251.167.101): icmp_seq=7 ttl=102 time=1.85 ms
64 bytes from ww-in-f101.1e100.net (142.251.167.101): icmp_seq=8 ttl=102 time=1.85 ms
64 bytes from ww-in-f101.1e100.net (142.251.167.101): icmp_seq=9 ttl=102 time=1.86 ms
64 bytes from ww-in-f101.1e100.net (142.251.167.101): icmp_seq=10 ttl=102 time=1.86 ms
64 bytes from ww-in-f101.1e100.net (142.251.167.101): icmp_seq=11 ttl=102 time=1.86 ms
64 bytes from ww-in-f101.1e100.net (142.251.167.101): icmp_seq=12 ttl=102 time=1.86 ms
64 bytes from ww-in-f101.1e100.net (142.251.167.101): icmp_seq=13 ttl=102 time=1.87 ms
64 bytes from ww-in-f101.1e100.net (142.251.167.101): icmp_seq=14 ttl=102 time=1.87 ms
```

CloudWatch

Favorites and recents

Dashboards

AI Operations

Alarms

Logs

Log groups

Log Anomalies

Live Tail

Logs Insights

Contributor Insights

Metrics

Application Signals (APM)

GenAI Observability

Network Monitoring

Insights

Log events

You can use the filter bar below to search for and match terms, phrases, or values in your log events.

142.251.167.101

Clear

1m

30m

1h

12h

Custom

Local timezone

Display

Timestamp	Message
2025-10-25T20:14:14.000-04:00	2 904727092191 eni-09ee9f1b54d81c064 142.251.167.101 10.0.1.92 0 0 1 6 504 1761437654 1761437678 ACCEPT OK
2 904727092191 eni-09ee9f1b54d81c064 142.251.167.101 10.0.1.92 0 0 1 6 504 1761437654 1761437678 ACCEPT OK	
2025-10-25T20:14:14.000-04:00	2 904727092191 eni-09ee9f1b54d81c064 10.0.1.92 142.251.167.101 0 0 1 6 504 1761437654 1761437678 ACCEPT OK
2 904727092191 eni-09ee9f1b54d81c064 10.0.1.92 142.251.167.101 0 0 1 6 504 1761437654 1761437678 ACCEPT OK	
2025-10-25T20:15:41.000-04:00	2 904727092191 eni-09ee9f1b54d81c064 10.0.1.92 142.251.167.101 0 0 1 6 5544 1761437741 1761437768 ACCEPT OK
2025-10-25T20:15:41.000-04:00	2 904727092191 eni-09ee9f1b54d81c064 142.251.167.101 10.0.1.92 0 0 1 6 5544 1761437741 1761437768 ACCEPT OK
2 904727092191 eni-09ee9f1b54d81c064 142.251.167.101 10.0.1.92 0 0 1 6 5544 1761437741 1761437768 ACCEPT OK	

CloudShell Feedback

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53°F Mostly cloudy

Search

ec2-user@ip-10-0-1-92:~

```
64 bytes from ww-in-f101.1e100.net (142.251.167.101): icmp_seq=445 ttl=102 time=1.86 ms
64 bytes from ww-in-f101.1e100.net (142.251.167.101): icmp_seq=446 ttl=102 time=1.85 ms
64 bytes from ww-in-f101.1e100.net (142.251.167.101): icmp_seq=447 ttl=102 time=1.86 ms
64 bytes from ww-in-f101.1e100.net (142.251.167.101): icmp_seq=448 ttl=102 time=1.87 ms
64 bytes from ww-in-f101.1e100.net (142.251.167.101): icmp_seq=449 ttl=102 time=1.88 ms
64 bytes from ww-in-f101.1e100.net (142.251.167.101): icmp_seq=450 ttl=102 time=1.87 ms
64 bytes from ww-in-f101.1e100.net (142.251.167.101): icmp_seq=451 ttl=102 time=2.09 ms
64 bytes from ww-in-f101.1e100.net (142.251.167.101): icmp_seq=452 ttl=102 time=1.93 ms
64 bytes from ww-in-f101.1e100.net (142.251.167.101): icmp_seq=453 ttl=102 time=1.87 ms
64 bytes from ww-in-f101.1e100.net (142.251.167.101): icmp_seq=454 ttl=102 time=1.92 ms
64 bytes from ww-in-f101.1e100.net (142.251.167.101): icmp_seq=455 ttl=102 time=1.90 ms
64 bytes from ww-in-f101.1e100.net (142.251.167.101): icmp_seq=456 ttl=102 time=1.85 ms
64 bytes from ww-in-f101.1e100.net (142.251.167.101): icmp_seq=457 ttl=102 time=1.85 ms
64 bytes from ww-in-f101.1e100.net (142.251.167.101): icmp_seq=458 ttl=102 time=1.91 ms
64 bytes from ww-in-f101.1e100.net (142.251.167.101): icmp_seq=459 ttl=102 time=1.91 ms
64 bytes from ww-in-f101.1e100.net (142.251.167.101): icmp_seq=460 ttl=102 time=1.84 ms
64 bytes from ww-in-f101.1e100.net (142.251.167.101): icmp_seq=461 ttl=102 time=1.87 ms
64 bytes from ww-in-f101.1e100.net (142.251.167.101): icmp_seq=462 ttl=102 time=1.85 ms
64 bytes from ww-in-f101.1e100.net (142.251.167.101): icmp_seq=463 ttl=102 time=1.88 ms
64 bytes from ww-in-f101.1e100.net (142.251.167.101): icmp_seq=464 ttl=102 time=1.86 ms
64 bytes from ww-in-f101.1e100.net (142.251.167.101): icmp_seq=465 ttl=102 time=1.86 ms
64 bytes from ww-in-f101.1e100.net (142.251.167.101): icmp_seq=466 ttl=102 time=1.87 ms
64 bytes from ww-in-f101.1e100.net (142.251.167.101): icmp_seq=467 ttl=102 time=2.02 ms
64 bytes from ww-in-f101.1e100.net (142.251.167.101): icmp_seq=468 ttl=102 time=1.93 ms
64 bytes from ww-in-f101.1e100.net (142.251.167.101): icmp_seq=469 ttl=102 time=1.85 ms
64 bytes from ww-in-f101.1e100.net (142.251.167.101): icmp_seq=470 ttl=102 time=1.86 ms
64 bytes from ww-in-f101.1e100.net (142.251.167.101): icmp_seq=471 ttl=102 time=1.88 ms
64 bytes from ww-in-f101.1e100.net (142.251.167.101): icmp_seq=472 ttl=102 time=1.85 ms
64 bytes from ww-in-f101.1e100.net (142.251.167.101): icmp_seq=473 ttl=102 time=1.86 ms
64 bytes from ww-in-f101.1e100.net (142.251.167.101): icmp_seq=474 ttl=102 time=1.87 ms
64 bytes from ww-in-f101.1e100.net (142.251.167.101): icmp_seq=475 ttl=102 time=1.92 ms
64 bytes from ww-in-f101.1e100.net (142.251.167.101): icmp_seq=476 ttl=102 time=1.84 ms
64 bytes from ww-in-f101.1e100.net (142.251.167.101): icmp_seq=477 ttl=102 time=1.86 ms
64 bytes from ww-in-f101.1e100.net (142.251.167.101): icmp_seq=478 ttl=102 time=1.86 ms
64 bytes from ww-in-f101.1e100.net (142.251.167.101): icmp_seq=479 ttl=102 time=1.86 ms
64 bytes from ww-in-f101.1e100.net (142.251.167.101): icmp_seq=480 ttl=102 time=1.86 ms
^C
--- google.com ping statistics ---
480 packets transmitted, 480 received, 0% packet loss, time 479664ms
rtt min/avg/max/mdev = 1.839/1.879/2.903/0.071 ms
[ec2-user@ip-10-0-1-92 ~]$
```

53°F Mostly cloudy Search 8:22 PM 10/25/2025

Shuting down EC2 Instance

The screenshot displays the AWS Management Console interface. At the top, the header shows the AWS logo, navigation icons, and account information: "United States (N. Vi)" and "Account ID: 9047-2709-2191" with the email "mlevin56-ENPM665-0101-2025".

The main content area is titled "VPC dashboard" and "Your VPCs". A green notification banner at the top states: "You successfully deleted vpc-08656f8506f5bf1c8 / MyLabVPC and 3 other resources." with a "Details" link.

Under "Your VPCs (1)", there is a search bar "Find VPCs by attribute or tag" and a table with one entry: "vpc-08656f8506f5bf1c8 / MyLabVPC". The table has columns for "Name" and "VPC ID".

The "Details" tab for the selected VPC shows the following information:

- VPC ID:** vpc-08656f8506f5bf1c8
- DNS resolution:** Enabled
- Main network ACL:** (link to details)

The left sidebar contains a navigation menu with categories: "Virtual private cloud" (including Subnets, Route tables, Internet gateways, Egress-only internet gateways, Carrier gateways, DHCP option sets, Elastic IPs, Managed prefix lists, NAT gateways, Peering connections, and Route servers) and "Security" (including Network ACLs).

The bottom of the console shows a footer with "CloudShell", "Feedback", "Privacy", "Terms", and "Cookie preferences". The system clock at the bottom right indicates "10:00 PM 10/25/2025".

Takeaways of This Assignment (Well-written reflection on security concepts, importance of VPC restrictions, and use of Flow Logs. Shows clear understanding of the lab.)

I learned how to build a secure VPC from start to finish. Each configuration step taught me how AWS controls traffic and access. When I created the subnet and route table, I understood how network boundaries are defined. Setting the security group to allow SSH from my IP helped me apply the rule of least privilege.

When I viewed the Flow Logs in CloudWatch, I didn't know how to filter only ICMP traffic, so I was stuck for a while. I had to figure out the IP address that I was pinging, which was 142.251.167.101 for google.com. Once I knew that, I could recognize the entries in the log and confirm that the pings were being captured. Later, I realized that when the protocol field value is 1, it means ICMP. That helped me understand exactly what kind of traffic I was seeing in the logs.

This lab helped me see that strong security in AWS depends on control. I control who connects, where traffic goes, and how it is logged. Building the VPC myself made me understand how isolation, routing, and visibility work together to keep a cloud environment secure.